

# **Geometric Encoded Medium**

The Vacuum Structured as a Bound Medium

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# 1

## Introduction

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### 1.0.0. The vision of Geometric Encoded Medium (GEM)

#### 1.0.1. Core Concept

GEM is a unified physical framework where all observable phenomena arise from exact geometric relationships encoded in a constrained vacuum substrate.

All Fields, forces, and particles emerge from the resonance of mass and charge within the medium's encoded structure.

#### 1.0.2. Motivation

Inspired by Nikola Tesla, I sought to gain a new perspective on the construct of the universe. I began investigations with light and all related quantities. Eventually I viewed light as a finite container, spherical and oscillatory in nature. Now, I view light as geometric encoding of action.

#### 1.0.3. Scope

Atomic energies, stellar acceleration, and black hole dynamics. Unifying scales in four dimensions which includes a dimension in the complex domain.

#### 1.0.5. Axiom

*Resonance is geometric action.*

#### 1.0.5. Principle

*All action is resonance encoded in a geometric medium.*

## 2

## Foundational Constants and the Bound Medium

### 2.0.0. Constants

#### 2.1.0. Intrinsic Constants of the Geometric Medium

##### 2.1.1. Speed of light

$$c = 299\,792\,458 \text{ [m/s]} \quad (2.1)$$

##### 2.1.2. Elementary charge

$$e = 1.602176634 \times 10^{-19} \text{ [C]} \quad (2.2)$$

##### 2.1.3. Planck constant

$$h = 6.62607015 \times 10^{-34} \text{ [J/Hz]} \quad (2.3)$$

##### 2.1.4. Geometric permeability encoding:

$$\Phi = 1 \times 10^{-7} \text{ [H/m]} \quad (2.4)$$

##### 2.1.5. Gravitational constant:

$$G = \frac{4\pi\Phi}{\Omega^2} = \frac{\pi^{3/4}}{25\,000\,000\,000\sqrt{2}} \left[ m^3 / (\text{kg s}^2) \right] \quad (2.5)$$

##### 2.1.6. Geometric mass-charge time-space encoding:

$$\Omega = 100 \sqrt[4]{4\pi} \left[ \frac{\text{kg s}}{\text{m C}} \right] \quad (2.6)$$

### 2.2.0. Derivation 1: Geometric Mass and Radius Coupling Constant

#### Theorem 1

The geometric vacuum constant  $\Gamma$ , coupling mass and radius, is equivalently expressible in terms of Planck-scale geometry and vacuum electromagnetism:

$$\Gamma = m_P l_P \alpha = e^2 \Phi \quad (2.7)$$

This identity unifies charge, mass, length, and vacuum response into a single encoded structure, foundational to all resonance phenomena in GEM.

### 2.3.0. Derivation 2: Fine Structure Constant

The fine structure constant's encoded parameters,  $\alpha_\gamma$  and  $\alpha_\delta$  emerged from geometrical compression scale:

$$\frac{M}{\Gamma} \left( R - \frac{2GM}{c^2} \right) = 1 \quad (2.8)$$

Where a mass is set as a n multiple Planck mass ( $m_P$ ),  $n \geq 1$ :

$$n = 1$$

$$M = n \cdot m_P = (1) m_P = m_P$$

$$\frac{m_P}{\Gamma} \left( R - \frac{2 G m_P}{c^2} \right) = 1 \Rightarrow R = \frac{l_P}{\Lambda_\alpha}$$

$$\frac{n \cdot m_P}{\Gamma} \left( \frac{l_P}{\Lambda_\alpha(n)} - \frac{2 G n \cdot m_P}{c^2} \right)$$

When R was solved for, a pattern was not revealed and the result looked arbitrary. So, I used  $l_P$ , Planck length, to scale R:

$$\frac{l_P}{R}$$

A pattern was revealed for normalization of R with  $l_P$ , a new expression symbolized as  $\Lambda_\alpha$ .

$$\frac{l_P}{R} \Rightarrow \Lambda_\alpha = \frac{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,000\,n}{3\,944\,089\,375\,000\,000\,000\,000\,000\,000\,000\,n^2 + 4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi}$$

### 2.3.1. Remarks

I noticed the numerator and denominator had “pieces” of the fine structure constant. the numerator of  $\Lambda_\alpha$  was the denominator of  $\alpha$  scaled by n, where n is the number of  $m_P$  used in the expression. And the denominator of  $\Lambda_\alpha$  showed something interesting; the left side was  $\frac{1}{2}$  the denominator of  $\alpha$  scaled by  $n^2$  and the right side of the denominator of  $\Lambda_\alpha$  was the numerator of  $\alpha$ .

From the Planck constants  $m_P$ ,  $l_P$  the gravitational constant  $G$ , the speed of light  $c$ , and the geometric encoding  $\Gamma$ , the fine structure constant  $\alpha$  is revealed.

Hence,

$$\alpha_\gamma = 4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi \quad (2.9)$$

$$\alpha_\delta = 1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,000 \quad (2.10)$$

$$\alpha = \frac{\alpha_\gamma}{\alpha_\delta} \quad (2.11)$$

(See **Table 3.1** ( $\alpha$ ) alignment with **CODATA** value)

$$\frac{1\,972\,044\,687\,500\,000\,000\,000\,000\,000\,000\,n}{3\,944\,089\,375\,000\,000\,000\,000\,000\,000\,000\,n^2 + 4\,580\,703\,784\,999\,263\,461\,548\,761\,\pi} \Rightarrow \frac{\alpha_\delta n}{2\,\alpha_\delta n^2 + \alpha_\gamma} = \Lambda_\alpha$$

$$\Lambda_\alpha = \frac{\alpha_\delta n}{2\,\alpha_\delta n^2 + \alpha_\gamma}, \text{ for } n \in \mathbb{Z} \quad (2.12)$$

### 2.4.0. Theorem 2: Geometric Modulation Identity

The geometric vacuum constant  $\Gamma$  is the fine-structure-scaled product of Planck mass and Planck length:

$$\Gamma = \alpha \cdot (m_P l_P)$$

### 2.5.0. Theorem 3

The geometric vacuum constant  $\Gamma$  relates directly to classical electron radius:

$$\Gamma = \frac{e^2 \mu_0}{4\pi} = \frac{e^2}{4\pi c^2 \epsilon_0}$$

This identity confirms that the classical radius is not arbitrary—it is a manifestation of encoded geometric response in the vacuum.

### 2.6.0. Derived Geometric Radii from $\Lambda_\alpha(n)$

In GEM, key quantum scales such as the classical radius, Compton wavelength, and Bohr radius emerge naturally from the Planck length via the scaling function  $\Lambda_\alpha(n)$ , where  $n$  is the Planck mass ratio of the system. These re-expressions reveal that all atomic structure is derived from geometric resonance.

■ Classical Particle Radius

$$r_m = \frac{l_P}{\Lambda_\alpha(n)} \quad (2.13)$$

■ Compton Wavelength

$$\lambda_c = \frac{l_P}{\Lambda_\alpha(n)} \frac{2\pi}{\alpha} \quad (2.14)$$

■ Bohr Radius

$$\alpha_0 = \frac{l_P}{\Lambda_\alpha(n)} \frac{1}{\alpha^2} \quad (2.15)$$

### 2.7.0. Example: Hydrogen Reduced Mass

$$\mu = \frac{m_e m_p}{m_e + m_p} \Rightarrow n = \frac{\mu}{m_p}$$

$$r_e = \frac{l_P}{\Lambda_\alpha(n)} = 2.819475019099183536025676295261 \times 10^{-15} \text{ m}$$

$$\lambda_c = \frac{l_P}{\Lambda_\alpha(n)} \frac{2\pi}{\alpha} = 2.427631645233921166912118719574 \times 10^{-12} \text{ m}$$

$$\alpha_0 = \frac{l_P}{\Lambda_\alpha(n)} \frac{1}{\alpha^2} = 5.294654093900544573540696115880 \times 10^{-11} \text{ m}$$

### 2.8.0. Vacuum Impedance and GEM Permittivity and Permeability Structure

#### 2.8.1. Theorem 4: Vacuum Impedance Scaling in GEM

$$Z_P = \frac{2h}{e^2} [\Omega] \quad (2.16)$$

$$\mu_P = \frac{1}{c} \cdot Z_P [\text{H/m}] \quad (2.17)$$

$$\epsilon_P = \frac{1}{c} \cdot \frac{1}{Z_P} [\text{F/m}] \quad (2.18)$$

#### 2.8.2. GEM-Scaled Emergence

$$Z_o = \alpha Z_P [\Omega] \quad (2.19)$$

$$\mu_o = \alpha \mu_P [\text{H/m}] \quad (2.20)$$

$$\epsilon_o = \frac{1}{\alpha} \epsilon_P [\text{F/m}] \quad (2.21)$$

$$\mu_o = G \Omega^2$$

### 2.9.0. VGEM Derivation of Rydberg Quantities

#### 2.9.1. Theorem 5: GEM Derivation of the Rydberg Series

■ Rydberg energy:

$$E_n = -\frac{\alpha^2}{2n^2} m c^2 \quad (2.22)$$

■ Rydberg frequency:

$$f_n = \frac{\alpha^2 m c^2 Z^2}{2 n^2 h} [\text{Hz}] \quad (2.23)$$

■ Rydberg wavelength:

$$\lambda_n = \frac{2 n^2 h}{Z^2 \alpha^2 m c} [\text{m}] \quad (2.24)$$

■ Rydberg angular frequency:

$$\omega_n = \frac{2 \pi \alpha^2 m c^2 Z^2}{2 n^2 h} [\text{rad/s}] \quad (2.25)$$

## 2.10.0. GEM Binding Energy Spectra Across Particles

■ Muon,  $(1.88353163 \times 10^{-28} \text{ kg})$ :

2.55927652  $\times 10^{-13}$  m 1.02371061  $\times 10^{-12}$  m 2.30334887  $\times 10^{-12}$  m 4.09484243  $\times 10^{-12}$  m 6.39819129  $\times 10^{-12}$  m  
 9.21339546  $\times 10^{-12}$  m 1.25404549  $\times 10^{-11}$  m 1.63793697  $\times 10^{-11}$  m 2.07301398  $\times 10^{-11}$  m 2.55927652  $\times 10^{-11}$  m  
 -2813.22580 eV -703.306451 eV -312.580645 eV -175.826613 eV  
 -112.529032 eV -78.1451612 eV -57.4127715 eV -43.9566532 eV -34.7311828 eV -28.1322580 eV

■ Proton,  $(1.67262193 \times 10^{-27} \text{ kg})$ :

2.88198916  $\times 10^{-14}$  m, 1.15279566  $\times 10^{-13}$  m, 2.59379025  $\times 10^{-13}$  m, 4.61118266  $\times 10^{-13}$  m, 7.20497291  $\times 10^{-13}$  m,  
 1.03751610  $\times 10^{-12}$  m, 1.41217469  $\times 10^{-12}$  m, 1.84447306  $\times 10^{-12}$  m, 2.33441122  $\times 10^{-12}$  m, 2.88198916  $\times 10^{-12}$  m  
 -24982.1298 eV, -6245.53245 eV, -2775.79220 eV, -1561.38311 eV, -999.285192 eV,  
 -693.948050 eV, -509.839384 eV, -390.345778 eV, -308.421356 eV, -249.821298 eV

■ Neutron,  $(1.67492750 \times 10^{-27} \text{ kg})$ :

2.87802204  $\times 10^{-14}$  m, 1.15120882  $\times 10^{-13}$  m, 2.59021984  $\times 10^{-13}$  m, 4.60483526  $\times 10^{-13}$  m, 7.19505510  $\times 10^{-13}$  m,  
 1.03608793  $\times 10^{-12}$  m, 1.41023080  $\times 10^{-12}$  m, 1.84193411  $\times 10^{-12}$  m, 2.33119785  $\times 10^{-12}$  m, 2.87802204  $\times 10^{-12}$  m  
 -25016.5657 eV, -6254.14142 eV, -2779.61841 eV, -1563.53535 eV, -1000.66263 eV,  
 -694.904602 eV, -510.542156 eV, -390.883838 eV, -308.846490 eV, -250.165657 eV

■ Tau,  $(3.16754000 \times 10^{-27} \text{ kg})$ :

1.52183659  $\times 10^{-14}$  m, 6.08734635  $\times 10^{-14}$  m, 1.36965293  $\times 10^{-13}$  m, 2.43493854  $\times 10^{-13}$  m, 3.80459147  $\times 10^{-13}$  m,  
 5.47861171  $\times 10^{-13}$  m, 7.45699928  $\times 10^{-13}$  m, 9.73975416  $\times 10^{-13}$  m, 1.23268764  $\times 10^{-12}$  m, 1.52183659  $\times 10^{-12}$  m  
 -47310.0910 eV, -11827.5227 eV, -5256.67677 eV, -2956.88068 eV, -1892.40364 eV,  
 -1314.16919 eV, -965.512060 eV, -739.220171 eV, -584.075197 eV, -473.100910 eV

■ Deuteron,  $(3.34300000 \times 10^{-27} \text{ kg})$ :

1.44196179  $\times 10^{-14}$  m, 5.76784716  $\times 10^{-14}$  m, 1.29776561  $\times 10^{-13}$  m, 2.30713886  $\times 10^{-13}$  m, 3.60490447  $\times 10^{-13}$  m,  
 5.19106244  $\times 10^{-13}$  m, 7.06561277  $\times 10^{-13}$  m, 9.22855545  $\times 10^{-13}$  m, 1.16798905  $\times 10^{-12}$  m, 1.44196179  $\times 10^{-12}$  m  
 -49930.7456 eV, -12482.6864 eV, -5547.86063 eV, -3120.67160 eV, -1997.22983 eV,  
 -1386.96516 eV, -1018.99481 eV, -780.167901 eV, -616.428959 eV, -499.307456 eV

■ Quark?,  $(5.52625196 \times 10^{-27} \text{ kg})$ :

8.72287094  $\times 10^{-15}$  m, 3.48914838  $\times 10^{-14}$  m, 7.85058385  $\times 10^{-14}$  m, 1.39565935  $\times 10^{-13}$  m, 2.18071774  $\times 10^{-13}$  m,  
 3.14023354  $\times 10^{-13}$  m, 4.27420676  $\times 10^{-13}$  m, 5.58263740  $\times 10^{-13}$  m, 7.06552547  $\times 10^{-13}$  m, 8.72287094  $\times 10^{-13}$  m  
 -82539.5994 eV, -20634.8999 eV, -9171.06660 eV, -5158.72496 eV, -3301.58398 eV,  
 -2292.76665 eV, -1684.48162 eV, -1289.68124 eV, -1019.00740 eV, -825.395994 eV

### 2.10.1. Summary

All quantum binding energy spectra are predicted precisely by scaling  $l_P$  and  $\alpha$  according to GEM geometry.

### 2.11.0. GEM Gravitational Scaling in Astronomical Systems

#### 2.11.1. Theorem 6: GEM Curvature and Gravitational Acceleration Emergence

$$d = \frac{m_P l_P}{\alpha} \cdot \frac{n^2}{m} \text{ [m]} \quad (2.26)$$

$$k = \sqrt[4]{1 - \frac{4 G m}{c^2 d}} \quad (2.27)$$

$$G_r = \frac{G}{k^2} \text{ [m}^2 \text{ / (kg} \cdot \text{s}^2 \text{)]} \quad (2.28)$$

3

Tables

Table 3.1. Constants

| Quantity                             | Units                          | GEM                           | Experimental Value<br>(CODATA) | Relative Error         |
|--------------------------------------|--------------------------------|-------------------------------|--------------------------------|------------------------|
| Fine Structure Constant ( $\alpha$ ) | const.                         | $7.2973525653 \times 10^{-3}$ | $7.2973525643 \times 10^{-3}$  | $2.2 \times 10^{-11}$  |
| Rydberg Constant                     | $\text{m}^{-1}$                | 0973731.57106                 | 10973731.568157                | $2.64 \times 10^{-10}$ |
| Gravitational Constant               | $\text{m}^3 / (\text{kg s}^2)$ | $6.67433 \times 10^{-11}$     | $6.67430 \times 10^{-11}$      | $1.81 \times 10^{-5}$  |
| Bohr Magneton Ratio                  | const.                         | -1.00116006085969             | -1.00115965218046              | $4.08 \times 10^{-7}$  |



## Appendix A: GEM Identities and Theorems Summary

This appendix collects the core identities, theorems, and geometric definitions used throughout the Geometric Encoded Medium (GEM) framework. These expressions serve as the backbone of all predictions, derivations, and interpretations presented in the main chapters.

### A.1 Core GEM Identities

#### A.1.1 Inverse Fine Structure Encoded

##### Vacuum Coupling Constant

$$\Gamma = \frac{e^2}{4\pi c^2 \epsilon_o} = \frac{e^2 \mu_o}{4\pi} = e^2 \Phi = \alpha m_P l_P \quad (\text{A.1})$$

##### Vacuum Impedance (Planck form):

$$Z_P = \frac{2h}{e^2} [\Omega] \quad (\text{A.2})$$

##### Vacuum Permeability:

$$\mu_P = \frac{1}{c} Z_P [\text{H/m}] \quad (\text{A.3})$$

##### Vacuum Permittivity:

$$\epsilon_P = \frac{1}{c} \frac{1}{Z_P} [\text{F/m}] \quad (\text{A.4})$$

#### A.1.2 Fine Structure Encoded

##### Vacuum Impedance:

$$Z_o = \alpha Z_P [\Omega] \quad (\text{A.5})$$

##### Vacuum Magnetic Permeability:

$$\mu_o = \alpha \mu_P [\text{H/m}] \quad (\text{A.6})$$

##### Vacuum Electric Permittivity:

$$\epsilon_o = \frac{1}{\alpha} \epsilon_P [\text{F/m}] \quad (\text{A.7})$$

## A.2 Theorems

### Theorem 1: Geometric Vacuum Coupling

The geometric vacuum constant  $\Gamma$  links charge, mass, and radius:

$$\Gamma = \frac{e^2}{4 \pi c^2 \epsilon_0} \quad (\text{A.8})$$

### Theorem 2: Geometric Modulation Identity

$\Gamma$  is the fine-structure-scaled Planck action:

$$\Gamma = \alpha m_P l_P \quad (\text{A.9})$$

### Theorem 3: Classical Radius Identity

$$r_e = \frac{\Gamma}{m_e} \quad (\text{A.10})$$

### Theorem 4: Vacuum Impedance Scaling

Vacuum field structure emerges from  $Z_P$  and  $\alpha$ :

$$Z_o = \alpha Z_P [\Omega] \quad (\text{A.11})$$

$$\mu_o = \alpha \mu_P [\text{H/m}] \quad (\text{A.12})$$

$$\epsilon_o = \frac{1}{\alpha} \epsilon_P [\text{F/m}] \quad (\text{A.13})$$

### Theorem 5: Rydberg Energy (GEM)

$$E_n = - \left( \frac{\alpha^2}{2 n^2} \right) m c^2 \quad (\text{A.14})$$

$$E_n = - \left( \frac{Z^2 \alpha^2}{2 n^2} \right) m c^2 \quad (\text{A.15})$$

### Theorem 6: Gravitational Scaling (GEM)

$$G_r = \frac{G}{k^2} [\text{m}^2 / (\text{kg} \cdot \text{s}^2)] \quad (\text{A.16})$$

$$k = \sqrt[4]{1 - \frac{4 G m}{c^2 d}} \quad (\text{A.17})$$

### Theorem 7: Geometric Unit Invariant

From the resonance structure:

$$\frac{V}{a} \frac{Q}{m_P^2} \frac{M c^2}{\alpha G} \Rightarrow \frac{\left( \frac{\alpha}{2\pi} \frac{c h M}{\Gamma Q} \right)}{\frac{\alpha}{2\pi} \frac{c h M^2}{\Gamma^2}} \frac{Q}{m_P^2} \frac{M c^2}{\alpha G} \Rightarrow \frac{c^2 \Gamma}{G m_P^2 \alpha} = 1 \quad (\text{A.18})$$

### Theorem 8: Anomalous Magnetic Moment (GEM)

The electron's anomalous magnetic moment emerges from GEM geometry:

$$\delta = \left( \frac{\alpha}{2\pi} - 1 \right) \frac{\alpha}{2\pi} \quad (\text{A.19})$$

### Theorem 9: Quantized Core Resonance

The limit where the medium can no longer stay real.

$$\kappa_m(r) = \sqrt[4]{1 - \frac{4 G m}{c^2 (r - l_P)}} \quad (\text{A.20})$$

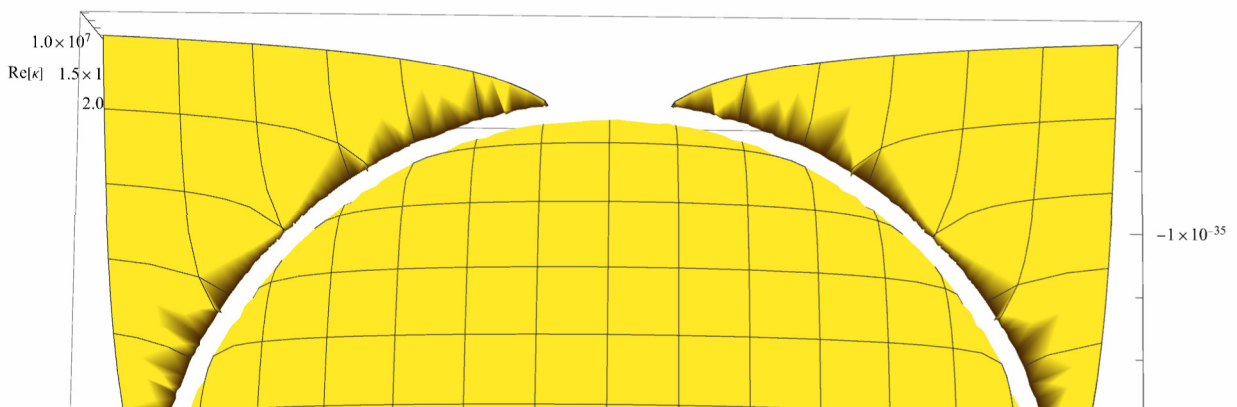
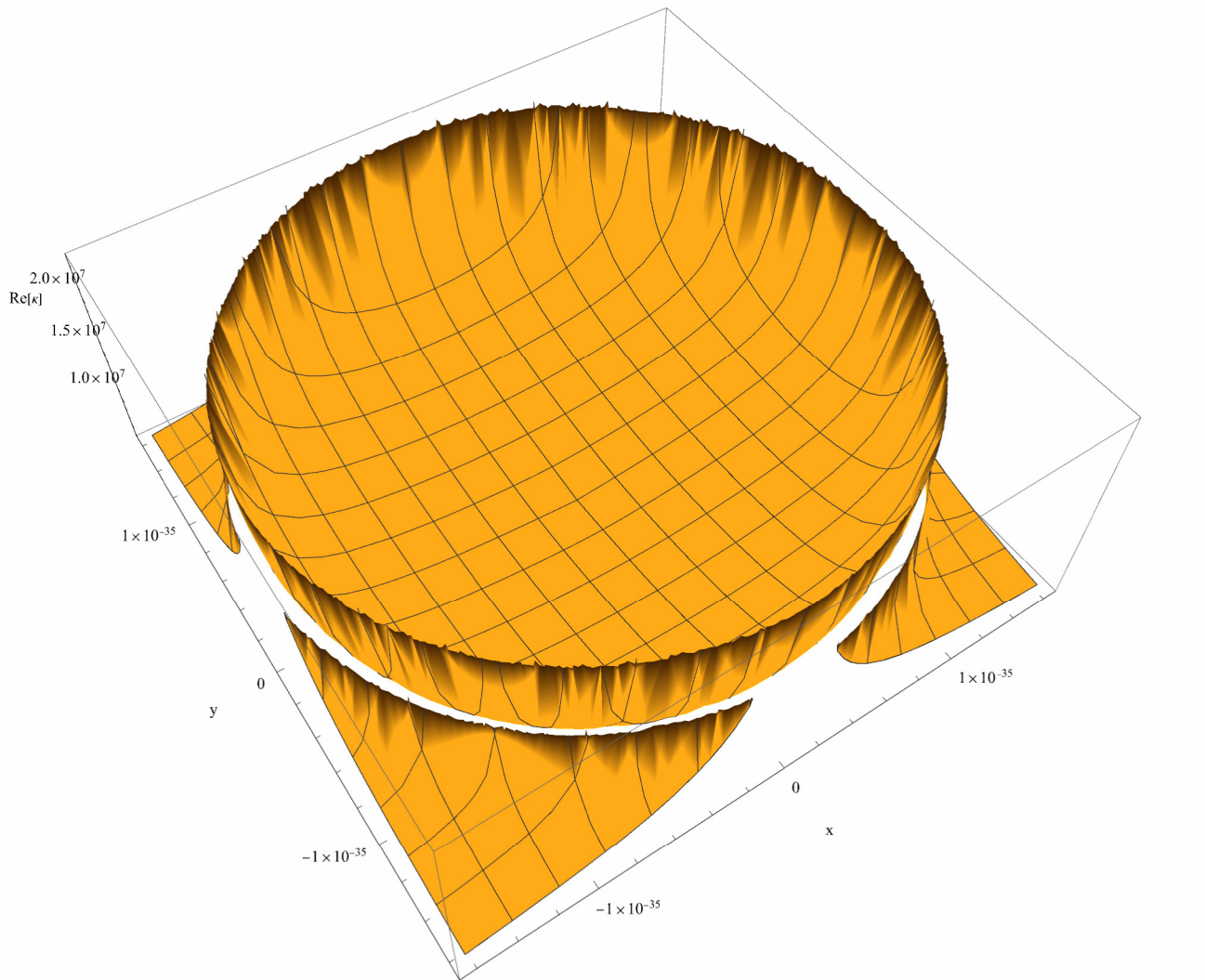
### A.3 Binding Energy Spectra

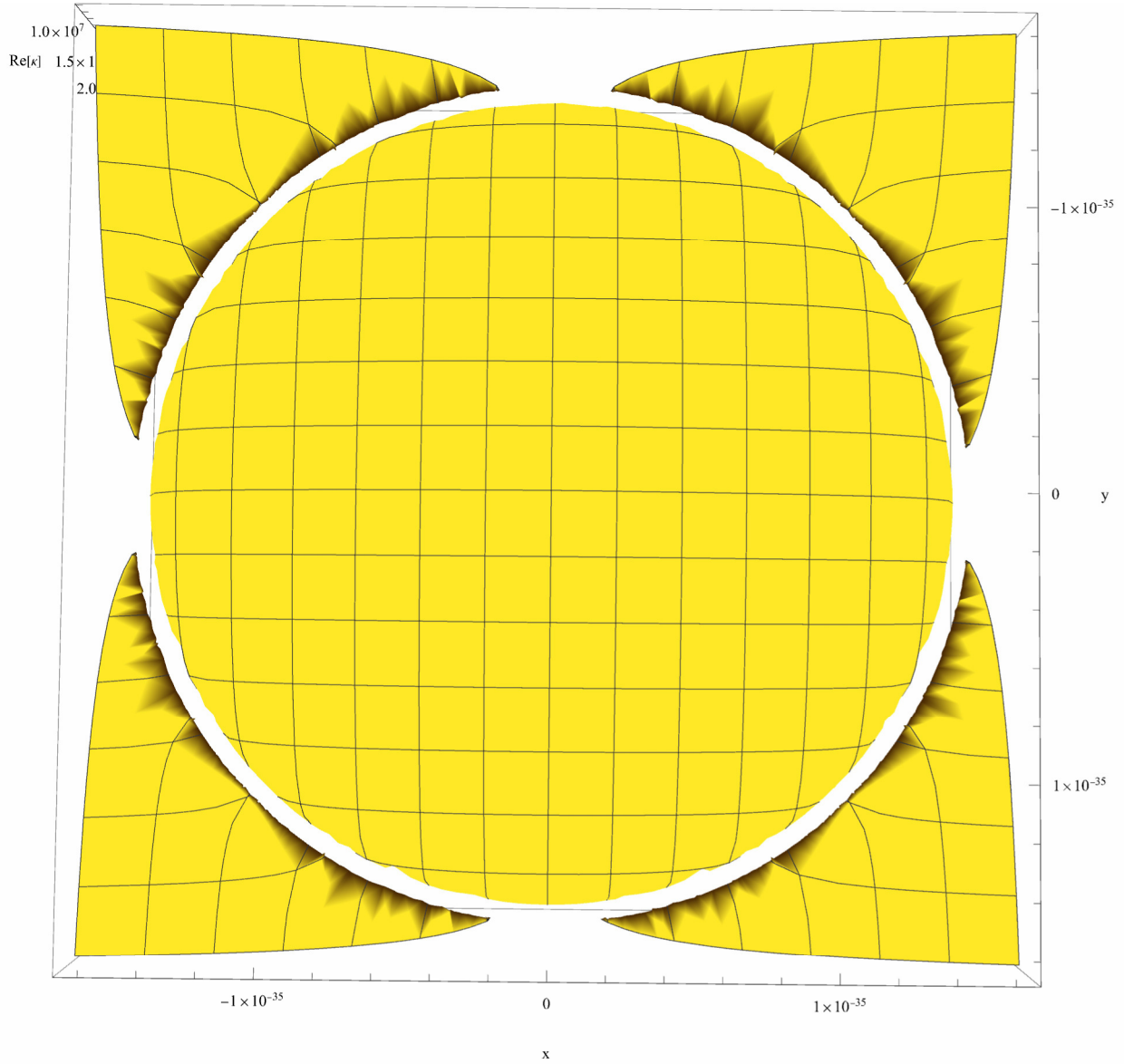
Muon, Proton, Neutron, Tau, Deuteron, see **Section 2.10.0.** for full data. Each follows:

$$E_n = -\left(\frac{\alpha^2}{2n^2}\right) m c^2$$

$$E_n = -\left(\frac{Z^2 \alpha^2}{2n^2}\right) m c^2$$

### Extras





$$V = \frac{\alpha \hbar c}{2\pi} \frac{M^2}{Q \Gamma^2} [\text{V}]$$

$$a = \frac{\alpha \hbar c}{2\pi} \frac{M^2}{\Gamma^2} [\text{m/s}^2]$$

$$a = \frac{\alpha \hbar c}{2\pi} \frac{M^2}{\Gamma^2} [\text{m/s}^2]$$

$$\frac{2 \sqrt{\alpha \hbar}}{e \sqrt{2 G c}} = \Omega$$

$$G \approx \frac{\mu_o}{\Omega^2} [m^3 / (\text{kg } s^2)]$$

$$\frac{2 \pi c e^2}{h} \frac{\mu_o}{4 \pi} = \frac{1}{4 \pi \epsilon_o} \frac{2 \pi c e^2}{h c^2} = \frac{e^2}{4 \pi \epsilon_o} \frac{2 \pi c}{h c^2} = \frac{e^2}{4 \pi \epsilon_o} \frac{2 \pi \sqrt{\mu_o \epsilon_o}}{h} = \alpha$$

$$E_n = -\frac{1}{n^2} \frac{m c^2 \pi \Gamma_o}{2 h^2} \frac{e^2 \Omega^2 G}{\mu_o \epsilon_o}$$

Example:

$$\mu = \frac{m_e m_p}{m_e + m_p} = 9.104425288916769957862728357084 \times 10^{-31} \text{ kg}$$

$$n = \frac{\mu}{m_p} = \frac{1}{23\,905\,189\,819\,168\,209\,745\,890}$$

$$r_e = \frac{1P}{\Lambda_\alpha[n]} = 2.819475019099183536025676295261 \times 10^{-15} \text{ m}$$

$$\lambda_c = \frac{1P}{\Lambda_\alpha[n]} \frac{2 \pi}{\alpha} = 2.427631645233921166912118719574 \times 10^{-12} \text{ m}$$

$$\alpha_\emptyset = \frac{1P}{\Lambda_\alpha[n]} \frac{1}{\alpha^2} = 5.294654093900544573540696115880 \times 10^{-11} \text{ m}$$

Equivalences

$$\frac{m_P}{M} = \frac{\alpha}{\pi M l_P} \left( \frac{h^2}{\mu_o c^2 e^2} \right) = \frac{c^2}{\sqrt{\frac{4 \pi \alpha c^2 M^2 G}{e^2 \mu_o}}} \quad (3.1)$$

Equivalences

$$\frac{m_P l_P}{m \alpha} n^2 = \frac{\alpha}{\pi M l_P} \left( \frac{h^2}{\mu_o c^2 e^2} \right) n^2 = \frac{c e l_P}{M \alpha \sqrt{\frac{4 \pi G}{\mu_o}}} n^2 \quad (3.2)$$

## Earth

Let  $m = 5.9722 \times 10^{24} \text{ kg}$  (Earth's mass)

Let  $r_{\text{Earth}} = 6.371 \times 10^6 \text{ m}$

$$R_s = \frac{2 G m}{c^2} \approx 8.87014 \times 10^{-3} \text{ m}$$

$$n = \sqrt{\frac{r_{\text{Earth}} m \alpha}{m_P l_P}} \approx 8.88436 \times 10^{35}$$

$$d = \frac{m_P l_P}{m \alpha} n^2 \approx r_{\text{Earth}}$$

$$k = \sqrt[4]{1 - \frac{4 G m}{c^2 d}} \approx 9.999999993038661827426949457731 \times 10^{-1}$$

$$G_r = \frac{G}{k^2} \approx 6.67432574113 \times 10^{-11} \text{ m}^3 / (\text{kg s}^2)$$

$$g = \frac{G_r m}{d^2} \approx 9.82034 \text{ m/s}^2$$

## Sun

Let  $m = 1.988400 \times 10^{30} \text{ kg}$  (Sun's mass)

Let  $r_{\text{Sun}} = 6.957 \times 10^8 \text{ m}$

$$R_s = \frac{2 G m}{c^2} \approx 2.95325 \times 10^3 \text{ m}$$

$$n = \sqrt{\frac{r_{\text{Earth}} m \alpha}{m_P l_P}} \approx 5.35695 \times 10^{39}$$

$$d = \frac{m_P l_P}{m \alpha} n^2 \approx r_{\text{Sun}}$$

$$k = \sqrt[4]{1 - \frac{4 G m}{c^2 d}} \approx 9.999978774930256871$$

$$G_r = \frac{G}{k^2} \approx 6.67435406453245 \times 10^{-11} \text{ m}^3 / (\text{kg s}^2)$$

$$g = \frac{G_r m}{d^2} \approx 2.74200972 \times 10^2 \text{ m/s}^2$$

### Neutron Star

Let  $m = 1.4 \times (1\,988\,400 \times 10^{24} \text{ kg}) = 2.78376 \times 10^{30} \text{ kg}$  (Neutron Star's mass)

Let  $r = 10 \times 10^4 \text{ m}$

$$R_s = \frac{2 G m}{c^2} \approx 4.13455 \times 10^3 \text{ m}$$

$$n = \sqrt{\frac{r_{\text{Earth}} m \alpha}{m_P l_P}} \approx 2.40309 \times 10^{37}$$

$$d = \frac{m_P l_P}{m \alpha} n^2 \approx r$$

$$k = \sqrt[4]{1 - \frac{4 G m}{c^2 d}} \approx 9.437471006418484131828 \times 10^{-1}$$

$$G_r = \frac{G}{k^2} \approx 7.4936971438876585 \times 10^{-11} \text{ m}^3 / (\text{kg s}^2)$$

$$g = \frac{G_r m}{d^2} \approx 2.08607 \times 10^{12} \text{ m/s}^2$$

# References

1. Authorlast, "Article Title," *Journal Title*, **Volume**(Issue), 2005 pp. #-#.\
2. Troy Deville. Geometric Encoded Medium (GEM): A Predictive Vacuum Framework Based on Planck Geometry. GitHub repository, <https://github.com/troydeville/Geometric-Encoded-Medium>, 2025.

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c= Quantity[299792458,  $\frac{\text{"Meters"}}{\text{"Seconds"}}$ ]
e = Quantity[1602176634×10-28, "Coulombs"]
h = Quantity[ $\frac{662607015}{10^{42}}$ ,  $\frac{\text{"Joules"}}{\text{"Hertz"}}$ ]
Ω=Quantity[100  $\sqrt[3]{4\pi}$ ,  $\frac{\text{"Kilograms" "Seconds"}}{\text{"Meters" "Coulombs"}}$ ]
αγ=4580703784999263461548761 π
αδ=19720446875000000000000000000
α=  $\frac{\alpha\gamma}{\alpha\delta}$ 
Γ=  $\frac{\alpha}{2\pi} \frac{h}{c}$ 
Φ=  $\frac{\Gamma}{e^2}$ 
ZP=UnitConvert[ $\frac{2}{e^2} \frac{h}{e^2}$ , "Ohms"]
μP=  $\frac{2h}{c e^2}$ 
εP=  $\frac{1}{2} \frac{e^2}{c h}$ 
Zo=ZP α
μO=μP α
εO=εP  $\frac{1}{\alpha}$ 
G=UnitConvert[ $\frac{4}{\Omega^2} \frac{\pi \Phi}{\text{"Meters"}^3}$ ,  $\frac{\text{"Kilograms" "Seconds"}^2}{\text{"Meters"}^3}$ ]
mP=  $\sqrt{\frac{h c}{2\pi G}}$  ;
lP=  $\sqrt{\frac{h G}{2\pi c^3}}$  ;
(*me=  $\frac{mP}{23892177732494625341423}$  *)

```

$$me = \text{UnitConvert}\left[\frac{mP}{23892177732494625341440}, \text{"Kilograms"}\right];$$

$$mp = \text{UnitConvert}\left[\frac{mP}{167262192595 \times 10^{-38}}, \text{"Kilograms"}\right];$$

$$(*mp = \frac{mP}{13012086673584404467} *)$$

$$m\mu = \frac{mP}{115550496527972941829}$$

$$\Delta\alpha[n_] := \frac{\alpha\delta \ n}{2\alpha\delta \ n^2 + \alpha\gamma};$$

$$rn[n_, m_] := \frac{mP \ 1P \ n^2}{\alpha \ m};$$

$$rn[n_, m_] := \frac{\Gamma \ n^2}{m \ \alpha^2};$$

$$\psi[n_] := \frac{n \ mP}{\Gamma} \left( \frac{1P}{\Delta\alpha[n]} - \frac{2G \ n \ mP}{c^2} \right);$$

$$\alpha == \frac{\Gamma \ 4\pi \ 2\pi \ c \ \Gamma}{e^2 \ \mu P \ h}$$

$$\alpha == \frac{2 \ \pi \ c \ e^2 \Phi}{h}$$

$$\frac{h \sqrt{2 \ \alpha}}{e \ \sqrt{c \ h \ 4\pi \ \Phi}} == 1$$

$$M = me$$

$$Q = -e$$

$$\text{UnitConvert}\left[\frac{\alpha \ c \ h \ M}{2\pi \ \Gamma}, \text{"Joules"}\right]$$

$$\text{UnitConvert}\left[\frac{\alpha \ c \ h \ M}{2\pi \ \Gamma \ Q}, \text{"Volts"}\right]$$

$$V = \frac{\alpha \ c \ h \ M}{2\pi \ \Gamma \ Q}$$

$$a = Q \frac{\alpha \ c \ h \ M^2}{2\pi \ \Gamma^2 \ Q}$$

$$a = \frac{\alpha \ c \ h \ M^2}{2\pi \ \Gamma^2}$$

$$\frac{V \ \Gamma \ Q}{a \ M \ \Gamma \ c^2}$$

$$\frac{V \ \Gamma \ Q \ c^2 \ \alpha\delta}{a \ M \ \Gamma \ G \ \alpha\gamma} == \frac{V \ \Gamma \ Q \ c^2}{a \ M \ \Gamma \ G \ \alpha}$$

$$\frac{\frac{V \ \Gamma \ Q \ c^2}{a \ M \ \Gamma \ G \ \alpha} \ mP}{\frac{mP}{M}} == \frac{mP}{M}$$

$$\frac{V}{a} \frac{\Gamma}{mP} \frac{Q}{\Gamma} \frac{c^2}{G \alpha} \frac{mP}{M} == \frac{mP}{M}$$
$$\frac{V}{a} \frac{Q}{mP} \frac{c^2}{G \alpha} \frac{mP}{M} == \frac{mP}{M}$$
$$\frac{V}{a} \frac{Q}{mP} \frac{c^2}{\alpha} \frac{M}{mP} == G$$
$$\frac{V}{a} \frac{Q}{mP} \frac{c^2}{\alpha} \frac{M}{G mP} == 1$$
$$\frac{V}{a} \frac{Q}{mP^2} \frac{M}{\alpha} \frac{c^2}{G} == 1$$
$$\frac{\frac{V}{a}}{\frac{\Gamma}{M} \frac{Q}{\Gamma} \frac{1}{c^2}} == \frac{1}{4\pi \epsilon_0}$$

3.